
Living in the IT Era

MST 101d Lecture Notes

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Preface

I am preparing these lecture notes for *Living in the IT Era*, taught to undergraduate biology students at Bulacan State University's San Rafael Campus. The course is not intended to turn students into computer scientists or to make them memorise catalogues of devices, inventors, and fashionable technologies. Its purpose is to help them understand the digital systems that shape ordinary life, use those systems effectively, evaluate information critically, protect themselves and others, and recognise when specialist technical knowledge is needed.

The material is organised into ten chapters. Each chapter is built around a central question and a practical task. Examples are drawn whenever possible from biology, health, research, and Philippine digital life.

The scope follows the general-education framework for *Living in the IT Era* in CHED Memorandum Order No. 20, series of 2013 [1]. The order places the course within the Mathematics, Science, and Technology elective domain and emphasises the science, culture, and ethics of information technology; its influence on modern life and relationships; and its use for personal, professional, and social advancement.

These notes are a work in progress. Chapters may be incomplete or revised as the course develops. Reports of errors and suggestions are welcome at

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I expect these notes to change in response to the students who use them. The first draft inevitably reflects what I know, what I can readily bring to mind, and what I expect a class will need. Students may identify a point that needs a better explanation, ask for a practical topic that I had not considered, or show that an apparently minor issue deserves more attention. I welcome such requests and am willing to revise the notes when doing so would make the course more useful.

That flexibility is deliberate. The module should respond both to the digital problems students face now and to those they are likely to meet later. My hope is that students will finish the course better prepared to understand new technologies, make sensible decisions about them, and meet the IT challenges ahead.

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Introduction

Most students already use information technology throughout the day. A phone serves as an alarm, camera, map, notebook, bank, library, entertainment centre, and means of communication. University work may involve search engines, learning platforms, shared documents, spreadsheets, online forms, biological databases, and AI assistants. Familiarity, however, is not the same as understanding. A person can use a system every day without knowing what it does with their information, why it sometimes fails, or what risks accompany its convenience.

This module is not intended to turn students into computer technicians. It aims to provide enough working knowledge to make better decisions: choosing a device or internet service, improving a home network, recognising a scam, protecting an identity, behaving responsibly and civilly online, understanding online tracking, using digital tools in scientific work, and judging claims about AI and other emerging technologies.

The organising idea is information. The usual definition of information technology can sound unnecessarily dry, but the central idea is simple: there is information, and IT is concerned with handling that information electronically. The following working definition is adapted from the University of Cincinnati's overview of the field [2]:

Information technology is the organised use of computers, networks, and software to create, process, store, communicate, and use electronic data.

This is deliberately broad. IT includes the hardware on which data are handled, the software that gives instructions to that hardware, and the networks that allow devices to exchange data. It also includes data management, cybersecurity, cloud computing, and newer applications such as artificial intelligence and machine learning [2]. These areas overlap. An online laboratory record, for example, may involve a phone or computer, a browser, a wireless network, a remote database, account permissions, and automated analysis.

IT matters because it changes what people can do with information. It can make communication faster, automate repetitive work, organise large collections of records, support decisions, and protect information from unauthorised access. In biology, these functions appear in field observations, laboratory instruments, sequence databases, statistical analysis, medical records, and collaboration between researchers. None of these systems removes the need for human judgement.

A computer can process a large data set quickly without knowing whether the measurements were sound or whether the conclusion is justified.

Data are recorded representations such as numbers, text, images, audio, and measurements. They become informative when they are interpreted in context. For example, 37.8 is only a recorded number until we learn that it is a patient's temperature in degrees Celsius. Context allows the value to answer a question or support a decision.

Most digital activities can be understood through five actions. A system receives **input**, performs some form of **processing**, may retain data in **storage**, produces **output**, and often **communicates** with another device. These actions need not occur in one machine. A student may enter information on a phone while the processing and storage take place in a distant data centre.

The Course at a Glance

The table below summarises the module. Later chapters examine each topic in greater depth. It is an outline, not a list of terms to memorise.

Table 1: The ten-chapter course map.

Chapter	Focus	What we shall examine
1	How IT works and why it matters	Information, computers, mobile devices, internet connections, and software choices as a working map of the module.
2	The physical foundations of computing	Chips, processors, materials, manufacturing, energy, water, data centres, and the supply chains behind apparently weightless digital activity.
3	The internet, Web, and home networks	How interconnected computers move information; the roles of internet service, routers, Ethernet, Wi-Fi, fibre, wireless links, undersea cables, cloud computing, mesh networks, and VPNs.
4	Software, cloud services, and digital work	Operating systems, applications, cloud storage, collaboration, file organisation, reproducibility, copyright, and the control exchanged for convenience.
5	Platforms, algorithms, and the information environment	Cookies, digital footprints, tracking, targeted advertising, recommendation systems, search, source evaluation, synthetic media, and AI-generated clutter.
6	Social media, attention, and relationships	Persuasive design, notifications, attention, social comparison, unrealistic expectations, disinhibition, harassment, and civility online.
7	Cybersecurity, identity, and cybercrime	Phishing, malicious links and downloads, scams, malware, peer-to-peer systems, identity theft, social engineering, and Philippine cybercrime law.

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Chapter	Focus	What we shall examine
8	Authentication, biometrics, privacy, and regulation	Passwords, multi-factor authentication, OTPs, passwordless methods, biometrics, privacy, the responsibilities created by collecting personal data, and the Philippine Data Privacy Act.
9	Artificial intelligence: how it works and fails	Machine learning and generative AI, including training, prediction, hallucination, cascading errors, bias, uncertainty, and sycophancy.
10	Responsible AI and the future of computing	Verification, privacy, copyright, academic integrity, human oversight, applications in biology and medicine, quantum computing, and post-quantum security.

Students are not expected to become specialists in every field listed here. They should learn enough to ask better questions, recognise when a claim requires checking, and know when expert help is needed.

How Classes Will Work. Class meetings will not simply repeat these notes. Every student is expected to read the assigned material before coming to class; the required pages or sections will be announced in advance. We shall use a **flipped classroom** approach: students first encounter the material through the assigned reading, while class time is used for a bird's-eye view of the chapter, questions, discussion, examples, and practical applications. Reading before class is therefore part of the course, not an optional preparation for it.

The assigned reading will not usually be terribly long. Much of the subject matter will already be familiar from everyday experience, although students may know one part from using a phone, another from browsing the Web, and another from hearing about online scams or AI. The notes bring these scattered pieces together, fill gaps, and provide a clearer account of how they are related. They are set in 12-point type with 1.2 line spacing. No one is being asked to read a dictionary from cover to cover.

Chapter 1

How IT Works and Why It Matters

Learning Outcomes

By the end of this chapter, students should be able to:

1. identify the basic components of a computer and explain their roles;
2. interpret selected laptop and mobile-phone specifications without assuming that larger numbers are always better;
3. distinguish common internet connection types and interpret download rate, upload rate, latency, jitter, and packet loss;
4. classify software by purpose, delivery, payment, and licence model; and
5. select hardware, an internet service, or software using criteria suited to the user's needs rather than marketing claims alone.

This chapter is a practical guide, not a catalogue to memorise. The tables are there to be consulted when a particular specification, connection, or software choice matters. In class, a few examples can be examined closely; the same questions can then be applied to devices and services that students actually encounter.

1.1 A Basic Map of a Computer

A computer is a collection of components with different jobs. The following table is intentionally basic; the objective is to recognise each component's role, not to memorise model numbers or become a repair technician.

Component	Main role	Plain-language description
Keyboard, mouse, camera, and microphone	Input	Allow a user or the environment to supply data to the computer.
CPU	General processing	Executes instructions and coordinates much of the computer's work. The CPU is a chip inside the computer, not the entire case.
GPU	Highly parallel processing	Produces graphics and performs many similar calculations at once; this is also useful in scientific computing and AI.
RAM	Working memory	Holds data and instructions that are being used. Its contents are normally lost when power is removed.
SSD (solid-state drive) or hard disk	Long-term storage	Retains the operating system, programs, documents, and other data when the computer is switched off.
Monitor and speakers	Output	Present visual or audio results to the user.
Ethernet and USB ports	Wired connection	Connect the computer to a network, storage device, peripheral, or other equipment.
Wi-Fi and Bluetooth	Wireless connection	Exchange data with networks or nearby devices without a cable.

The components work together. Pressing a key creates input; software assigns meaning to it; the CPU processes the instruction using data held in RAM; the result may appear on the monitor, be saved to storage, or be transmitted over a network. A more powerful component is useful only when the rest of the system and the intended task can take advantage of it.

1.2 Reading Hardware Specifications

The two examples below are expensive flagship devices because their specification sheets contain many of the terms that buyers encounter. They are not models that students are expected to own. The same method applies to a modest laptop or phone: begin with the work it must do, then ask which figures matter for that work and which are mainly selling points.

1.2.1 Reading a Laptop Specification Sheet

A laptop specification sheet compresses a great deal of information into a small space. Before interpreting the terms, let us organise one actual example. Table 1.1 summarises the supplied specification sheet for a 2026 16-inch MacBook Pro [3]. The machine is a high-end example, not a recommendation or a minimum requirement for students.

Table 1.1: Selected specifications from the supplied laptop specification sheet.

Specification	Value listed in the sheet
Model and identifiers	Apple MacBook Pro, 16-inch, M5 Max (2026); model number A3429; model identifier Mac17,6
Processor (CPU)	Apple M5 Max with 18 CPU cores; Geekbench 6 scores of 4,307 for single-core and 28,978 for multi-core performance
Graphics (GPU) and AI hardware	Integrated 40-core Apple GPU and 16-core Neural Engine
Memory (RAM)	48 GB unified memory as standard; configurable to 64 GB or 128 GB when purchased; soldered and not later upgradeable
Storage	2 TB built-in SSD as standard; 4 TB or 8 TB available when purchased; not later upgradeable
Built-in display	16.2-inch Liquid Retina XDR display; 3456 by 2234 pixels; 254 pixels per inch; adaptive refresh rate up to 120 Hz
Camera and audio	12-megapixel Centre Stage camera with 1080p video and a six-speaker sound system
Wireless connections	Wi-Fi 7, Bluetooth 6, and Thread
Physical connections	Three Thunderbolt 5 ports using USB-C connectors, HDMI, an SDXC card slot, a 3.5 mm headphone socket, and MagSafe 3 charging; no built-in Ethernet port
External displays	Support for as many as four external displays, subject to limits on their resolution and refresh rate
Battery and charging	100 Wh battery; claimed maximum of 16 hours of wireless web use or 22 hours of video streaming; 140 W power adaptor
Operating system	macOS 26.3.1 originally installed; the sheet lists no native Windows support and points instead to virtualisation
Dimensions and mass	0.66 by 14.01 by 9.77 inches; 4.7 lb (2.14 kg)
Price in the source	Original price of US\$4,399; estimated retail price of US\$4,999 at the time represented by the sheet

The model name tells us which machine is being described, but identifiers such as A3429 and Mac17,6 are often more reliable when checking compatibility, replacement parts, or support. Two laptops sold under similar names may belong to different generations. The date and exact identifier therefore matter.

Processor, graphics, and memory. A CPU's core count and clock speed do not by themselves establish that it is faster than another CPU. The processor generation, design, cooling, and task also matter. A **benchmark** is a standardised performance test. Its results can help, but only when the same benchmark and version are used under comparable conditions. A single-core score reflects work that depends mainly on one core; a multi-core score reflects work that can be divided among several cores. GPU core counts are likewise poor for comparisons across different designs

or manufacturers. They become meaningful only with suitable tests of the work one actually intends to perform. The Neural Engine is Apple's name for hardware designed to perform certain machine-learning calculations.

RAM is the computer's short-term working space. More RAM allows more programs and larger data sets to remain readily available, but unused capacity does not make every task faster. In this machine the CPU and GPU share unified memory. The important purchasing detail is that the memory is soldered: its capacity must be chosen at purchase because it cannot be expanded later.

Storage and upgradeability. The 2 TB figure describes how much the built-in SSD can store, not its speed or the amount of RAM. Photographs, videos, large data sets, and applications consume storage over time. This laptop's internal storage cannot be upgraded after purchase, although external drives and network or cloud storage remain possible. A buyer should distinguish among the standard configuration, an option available when ordering, and a component that can be upgraded later; these are not the same thing.

Display and connections. Screen size is measured diagonally. Resolution gives the number of pixels, pixel density describes how closely they are packed, and refresh rate gives the maximum number of times the display can update each second. Higher values can improve particular uses, but they may also increase cost or power consumption. The software being used must be able to benefit from them.

Port descriptions require careful reading. USB-C describes the connector's shape, whereas Thunderbolt 5 describes capabilities such as data transfer, display output, and charging. Devices with the same connector need not support the same functions or speeds. The absence of built-in Ethernet does not prevent a wired network connection, but it requires a compatible adaptor or dock. Likewise, obtaining the benefits of Wi-Fi 7 requires a compatible network, not merely a compatible laptop.

Battery, portability, and software. The words "up to" mark a best-case claim measured under stated test conditions. Screen brightness, wireless activity, software, temperature, and workload can make actual battery life shorter. Mass and dimensions matter when a laptop is carried daily, while the operating system matters because required applications and laboratory equipment may support only particular platforms. A powerful computer that cannot run the required software is the wrong computer for the task.

From specifications to a sensible choice. Begin with the work the computer must perform, the software and connections it requires, the need for portability, and the available budget. Then

compare machines using consistent definitions and suitable tests. Specifications cannot fully describe keyboard comfort, fan noise, repairability, reliability, warranty service, or real-world battery life, so independent reviews and first-hand inspection remain useful. The best choice is not necessarily the machine with the largest numbers; it is the one that meets the user’s needs at an acceptable total cost.

The source itself also illustrates why specifications should be checked. It reports two different memory-bandwidth figures, 614 GB/s and 460 GB/s. Apple’s specification resolves the discrepancy: the 40-core GPU configuration in this example has 614 GB/s of memory bandwidth, while 460 GB/s belongs to the 32-core GPU configuration [4]. The two figures therefore describe different configurations; only the first applies to the machine in the table. A polished table can organise claims, but it does not make them correct.

1.2.2 Reading a Mobile-Phone Specification Sheet

Mobile-phone specification sheets use many of the same terms as laptop sheets, but add cellular networks, SIMs, cameras, positioning, sensors, and resistance to water and dust. Table 1.2 organises selected information from the supplied specification sheet for the Apple iPhone 17 Pro Max [5]. This is again a worked example rather than a recommendation.

Table 1.2: Selected specifications from the supplied mobile-phone specification sheet.

Specification	Value listed in the sheet
Model and release	Apple iPhone 17 Pro Max; announced 9 September 2025 and released 19 September 2025
Regional versions	A3526 (international), A3257 (United States), A3525 (Middle East, Canada, and Mexico), and A3527 (China)
Mobile networks	GSM, CDMA, HSPA, EVDO, LTE, and 5G
SIM arrangement	Varies by market: combinations include Nano-SIM and eSIM, eSIM only, or two Nano-SIMs
Processor and graphics	Apple A19 Pro chipset; six-core CPU and six-core Apple GPU
Memory and storage	12 GB RAM; 256 GB, 512 GB, 1 TB, or 2 TB of built-in NVMe storage; no memory-card slot
Display	6.9-inch LTPO Super Retina XDR OLED; 1320 by 2868 pixels; about 460 pixels per inch; refresh rate up to 120 Hz; HDR10 and Dolby Vision; advertised brightness of 1,000 nits typical, 1,600 nits in high-brightness mode, and 3,000 nits peak; GSMArena measured 1,000 nits
Rear cameras	Three 48-megapixel cameras: wide, periscope telephoto with four-times optical zoom, and ultrawide; LiDAR depth scanner
Front camera	18-megapixel camera with an additional depth and biometrics sensor

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Specification	Value listed in the sheet
Video	Rear-camera recording up to 4K at 120 frames per second, with available rates and formats depending on the selected mode
Wireless and positioning	Wi-Fi 7, Bluetooth 6.0, NFC, dual-frequency GPS and other satellite-positioning systems; selected satellite communication features
Physical connection	USB Type-C 3.2 Gen 2 with DisplayPort; no 3.5 mm headphone socket
Sensors and authentication	Face ID, accelerometer, gyroscope, proximity sensor, compass, barometer, and second-generation Ultra Wideband support
Battery	4,823 mAh for the Nano-SIM model or 5,088 mAh for the eSIM-only model; GSMArena active-use test result of 17 hours and 58 minutes
Charging	Wired USB Power Delivery; claimed 50 per cent charge in 20 minutes; up to 25 W wireless MagSafe or Qi2 charging; 4.5 W reverse wired charging
Construction and resistance	Aluminium frame; Ceramic Shield glass; IP68 dust and water resistance, with the sheet specifying immersion to 6 m for 30 minutes
Dimensions and mass	The supplied sheet lists 163.4 by 78 by 8.8 mm and 233 g; Apple's Philippine specification lists 163.4 by 78 by 8.75 mm and 231 g [6]
Operating system	iOS 26 in the listed configuration, described in the sheet as upgradeable to iOS 26.6
Performance tests	AnTuTu scores of 2,600,583 in version 10 and 2,331,417 in version 11; Geekbench 6 score of 10,118; 3DMark Wild Life Extreme score of 6,010
Price in the source	Listed from approximately US\$1,186, with prices varying by storage capacity, seller, market, and date

Start with the exact version. The product name is not enough. The model number identifies the regional version, which can affect the available SIM arrangement, battery capacity, network compatibility, and warranty service. A phone imported from another market may therefore differ from the version sold locally even when both carry the same product name. Before buying, check the exact model against the mobile operator's supported frequency bands and the manufacturer's local support terms. A label such as "5G" does not guarantee 5G service everywhere: the phone, network bands, operator, subscription, and local coverage must all be compatible. A Nano-SIM is a small physical card, whereas an eSIM is a digital SIM profile stored by the phone.

Processor, RAM, and storage. As with laptops, CPU and GPU core counts are meaningful only within the context of a particular design. Tests of real applications are more informative than a core count alone. The table also shows why benchmark versions must not be mixed: the same phone receives different AnTuTu scores under versions 10 and 11. This does not mean that its hardware changed between the two tests.

RAM supports programs while they are running; storage retains applications, photographs, videos, and other files. The storage capacity must be chosen at purchase because this phone has no memory-card slot. Cloud storage can provide additional space and synchronisation, but it may

require a subscription and an internet connection and should not be confused with increasing the phone's built-in storage.

Display and cameras. Display size is measured diagonally, while resolution, pixel density, refresh rate, brightness, colour reproduction, and power use describe different properties. The sheet lists both advertised brightness levels and a separately measured result. Such figures should be compared only when the testing conditions and definitions match.

A camera's megapixel count describes the number of recorded pixels, not the overall quality of every photograph. Sensor size, lens, aperture, optical image stabilisation, focusing, image processing, lighting, and the photographer all matter. Multiple rear cameras provide different fields of view or optical capabilities; three 48-megapixel cameras should not be read as a single 144-megapixel camera. Optical zoom uses the camera's optics, whereas digital zoom enlarges or crops the recorded image and may lose detail.

Battery, charging, and durability. The two battery capacities in the table refer to different regional versions, not contradictory measurements of one battery. Capacity in milliampere-hours is useful, but it cannot by itself predict running time across phones with different displays, processors, software, and network conditions. A controlled active-use test is generally more informative, although no single test can reproduce every person's use. Fast-charging times are also conditional on a suitable charger, cable, temperature, battery state, and charging protocol.

An IP68 rating indicates performance under specified laboratory conditions; it does not mean that the phone is impossible to damage with liquid. Wear, impact, repairs, depth, time, temperature, pressure, and the type of liquid can affect resistance. Durability, repairability, spare-parts availability, and warranty conditions deserve attention alongside processor and camera figures.

Software, privacy, and value. The operating-system version listed on a specification sheet is a snapshot, not a promise of support for a particular number of years. Check the manufacturer's current update policy and whether essential applications will run on the device. Face ID is convenient, but biometric authentication also raises questions about how biometric data are stored and what fallback method is used; these questions return in our study of privacy and cybersecurity.

Price is unusually unstable: it changes with storage, market, taxes, seller, promotions, and time. A sensible comparison fixes the required storage and features, checks the exact regional model, and considers the total cost of the phone, accessories, repairs, cloud services, and expected period of use. The best phone is the one that meets those requirements reliably, not simply the one with the longest specification sheet.

For biology students, a phone may also be a field notebook, camera, location recorder, timer, reference tool, or means of entering observations. Suitability for scientific work depends not only on the hardware but also on measurement accuracy, consistent methods, file formats, data export, privacy, and whether the work can be checked or repeated.

1.3 Internet Connection Types

A useful device still needs a suitable connection. The internet reaches homes and other premises through several technologies. The connection type affects the speeds that are possible, the consistency of those speeds, latency, reliability, installation, and cost. It also determines which services can feasibly be offered at a particular address. The following summary draws on the supplied overview of home-internet technologies but omits its provider-specific and United States market information [7].

Connection	How data reach the premises	Main characteristics
Fibre-optic	Pulses of light travel through thin strands of glass or plastic.	Can provide very high and often similar download and upload speeds. Building the required network is costly, so service is not available everywhere.
Cable	Electrical signals travel through coaxial cable of the kind also used for cable television.	Can provide high download speeds, but upload speeds are often lower. Capacity may be shared locally, so performance can fall during busy periods.
DSL	Electrical signals travel through copper telephone lines.	Makes use of existing telephone infrastructure and, unlike dial-up, allows telephone voice service to be used at the same time. Performance depends strongly on line quality and distance from network equipment and is generally more limited than fibre or cable.
Fixed wireless	Radio signals travel between a base station and a receiver at a fixed location.	Can serve places where laying a cable is impractical. Distance, obstructions, alignment, interference, and network demand can affect performance; some systems require a mounted outdoor receiver.
Cellular home internet	A router receives a 4G LTE or 5G mobile-network signal and shares the connection with devices in the premises.	Is often quick to install, but performance varies with coverage, indoor signal strength, network load, and the router's position.
Satellite	A dish or terminal exchanges radio signals with satellites in orbit.	Can reach remote places without nearby terrestrial cables or towers. It requires a suitable view of the sky and may involve costly equipment, data limits, and greater latency. Low-Earth-orbit systems reduce the signal's travel distance compared with geostationary systems.

In the broad technical sense, cellular home internet is also a form of fixed wireless access. The table separates it because consumer plans based on mobile 4G or 5G networks are commonly discussed apart from systems using a dedicated fixed receiver.

These descriptions identify broad families rather than guaranteeing a particular level of service. Two connections using the same technology may perform differently because of the local infrastructure, distance, equipment, congestion, maintenance, and plan conditions. Parts of a network may also use different media. For example, fibre may carry data across most of a provider's network while another technology completes the final connection to a building. The exact connection installed at the premises matters more than a vague marketing label.

The internet connection is not the Wi-Fi. The service provider's connection brings internet access to a modem, optical network terminal, fixed receiver, satellite terminal, or cellular router. An optical network terminal is the box at the end of a fibre connection. Wi-Fi is the short-range wireless network that a router normally uses to share the connection within the premises. A weak Wi-Fi signal in one bedroom does not necessarily mean that the internet service entering the home is slow. Conversely, an excellent Wi-Fi signal cannot make a slow external connection faster. Testing a computer over a wired Ethernet connection, where practicable, helps distinguish an internal Wi-Fi problem from a problem with the internet service itself.

What the performance figures mean. Download rate describes how quickly data travel from the internet to the user, as when loading a page or streaming a video. Upload rate describes travel in the other direction, as when sending a file, backing up photographs, joining a video call, or submitting a large data set. A connection is **symmetrical** when its download and upload rates are approximately equal; otherwise it is **asymmetrical**.

Connection rates are normally advertised in megabits per second (Mbps) or gigabits per second (Gbps), while file sizes are commonly stated in megabytes (MB) or gigabytes (GB). Since one byte contains eight bits, a rate of 100 Mbps corresponds to at most

$$\frac{100}{8} = 12.5 \text{ MB/s}$$

before overhead and other real-world limitations are considered. Here, **overhead** means the additional data and work required to manage the transmission itself. Thus, Mbps and MB/s are not interchangeable. The time required to transfer a file can be estimated by dividing its size in bits by the transfer rate in bits per second, provided that the units are first made consistent.

Rate is not the whole story. **Latency** is the time taken for data to travel between points in a network. Speed tests commonly report the round-trip time in milliseconds; a smaller value is generally better for conversation, remote control, and online games. **Jitter** is variation in that delay, while **packet loss** occurs when some transmitted data fail to arrive and may have to be

recovered or sent again. A connection with a high advertised rate can still feel poor during a video call if latency, jitter, or packet loss is excessive.

Choosing a connection and plan. Begin with what is actually available at the exact address. Consider how many people and devices will use the connection at once, and distinguish download-heavy activities from tasks that need substantial upload capacity or low latency. Read whether an advertised rate is a maximum or an expected range and whether the service has a data allowance, a fair-use policy, or reduced speeds after a threshold. Account for installation charges, required equipment, contract length, post-promotional price, repair response, and the cost of ending the service.

When comparing plans in the Philippines, give the provider the exact service address and ask which connection will actually be installed. Availability can differ within the same municipality or city, and the advertised plan may not be offered at every address.

Once connected, test at different times and record download rate, upload rate, latency, and whether the test used Ethernet or Wi-Fi. A single result is only a snapshot. Several measurements under stated conditions provide better evidence of the service the household actually receives.

1.4 Software Selection

Hardware determines what a device is physically capable of doing, but software turns that capacity towards a particular task. Choosing software therefore begins with the task, not with whichever application happens to be popular. Students must also consider compatibility, cost, privacy, security, support, and whether they will still be able to open or export their work later.

1.4.1 Types of Software by Use

Software can be grouped according to its principal purpose. These categories are convenient rather than absolute: one application may perform several kinds of work and therefore belong to more than one category.

Table 1.3: Common types of software classified by use.

Type	Principal use	Examples of tasks
System software	Operates the device and manages its hardware resources.	Running the operating system, controlling a printer, and managing files and memory.
Productivity software	Helps users create, organise, calculate, and present work.	Writing documents, using spreadsheets, preparing presentations, and taking notes.

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Type	Principal use	Examples of tasks
Communication and collaboration software	Allows people to exchange messages or work together.	Email, instant messaging, video meetings, shared calendars, and collaborative editing.
Scientific and data software	Collects, manages, analyses, models, or visualises data.	Statistical analysis, database work, programming, bioinformatics, mapping, and laboratory data collection.
Creative software	Produces or edits visual, audio, and multimedia material.	Drawing, photography, page layout, animation, audio recording, and video editing.
Utility software	Maintains, organises, diagnoses, or extends a computer system.	Backing up data, compressing files, managing storage, and monitoring system performance.
Security software	Helps protect devices, accounts, networks, or data.	Malware detection, password management, encryption, firewalls, and secure authentication.
Educational software	Supports teaching, learning, practice, or assessment.	Learning platforms, simulations, tutorials, reference tools, and quizzes.
Entertainment software	Provides recreation or access to media.	Games, media players, streaming, and digital reading.

The intended use determines which features deserve attention. A spreadsheet may be entirely adequate for a small table but awkward or unsafe for a large, shared research data set. A sophisticated statistical package may be powerful but unnecessarily difficult for a simple calculation. Software should be proportionate to the task and the consequences of error.

1.4.2 How Software Is Supplied and Paid For

The labels used for software do not all describe the same thing. Some describe how software is delivered, some describe how payment works, and others describe what its licence permits. A single product may therefore carry several labels. For example, an online service can be SaaS, subscription-funded, and partly open source at the same time. **Proprietary software** is software whose owner does not normally supply the source code or grant users permission to inspect, modify, and redistribute it.

Table 1.4: Common ways in which software is supplied, licensed, or funded.

Model	What it means	Important qualification
Perpetual or one-time-purchase licence	The user pays once for permission to use a particular product or version indefinitely.	The payment may not include future major versions, continuing support, cloud storage, or compatibility with future devices. The user normally buys a licence, not ownership of the code.
Subscription licence	Continued use or access requires payment at regular intervals.	Updates and services may be included, but access to the software or some features may end when the subscription ends.
Software as a Service (SaaS)	The provider operates the software as an online service, commonly accessed through a browser or application.	SaaS describes delivery, not necessarily price. It may be subscription-funded, free to the user, advertising-funded, or paid for by an institution. Internet access and the provider's continued operation may be essential.
Freeware	Proprietary software is supplied without a monetary charge.	A zero price does not normally grant permission to inspect, modify, or redistribute the source code. Other costs may include advertising, data collection, or paid companion services.
Free and open-source software (FOSS)	The source code is available under a licence that grants rights to use, inspect, modify, and redistribute it subject to stated conditions.	Open source does not necessarily mean that every service, copy, or form of support is free of charge. Its licence obligations still apply.
Freemium	A basic version is free, while additional features, capacity, or convenience require payment.	The free version may impose storage, export, collaboration, usage, or advertising limits.
Institutional or educational access	A university, school, or employer pays for or negotiates access on behalf of eligible users.	Access may end when the agreement expires or the user leaves the institution. Personal files should be exported before eligibility ends.
Trial or demonstration version	Use is allowed temporarily or with restricted features so that the product can be evaluated.	The software may stop working, limit saving or export, or require payment after the trial.

These arrangements should not be confused with the distinction between software installed on a device and software operated remotely. Installed software can require a subscription, while a web-based service can be provided without charge. Nor should “freeware” be used as a synonym for “open source”: the first concerns price, whereas the second concerns licence permissions.

1.4.3 Choosing Software

Begin by stating what the software must accomplish and which file formats, devices, operating systems, and collaborators it must support. Test whether it can import existing work and export data in a form that other applications can read. A convenient tool can become a trap if all of the user's work is held in a proprietary format that cannot be moved elsewhere.

The relevant price is the total cost over the expected period of use. This may include renewals, storage, additional users, paid extensions, training, accessories, and the cost of moving data to another system. A one-time purchase is not automatically cheaper than a subscription, nor is free software automatically the better choice. The answer depends on the task and the period over which the software will be needed.

Privacy and security matter whenever software handles personal, academic, financial, health, or research data. Before using it, check what data it collects, where those data are stored, who can access them, whether suitable backups and encryption are available, and how the provider responds to security problems. Software should come from a dependable source and receive appropriate updates. Open source permits inspection of the code, but that fact alone does not guarantee that a particular program is secure or well maintained.

Also consider usability, accessibility, documentation, technical support, offline operation, and the ability of collaborators to use the same files. Using an unlicensed copy of commercial software is not a sensible way to save money. An altered installer may contain malicious software, and unauthorised use also breaches the licence. A simpler legal tool that does the required job is usually the better choice.

Freeware and paid software. Freeware removes the immediate purchase cost and can make useful tools available to people who could not otherwise afford them. It may also be convenient for a small or occasional task for which buying a large commercial package would be excessive. Many freeware programs come from reputable developers and are maintained responsibly. The absence of a price is not itself a warning that software is malicious.

It is equally important not to assume that every free download is safe. A file presented as freeware may have been altered, repackaged, or imitated by someone other than the real developer. Some freeware is supported by advertising, optional bundled programs, extensive data collection, or repeated pressure to buy another product. The program itself may not contain malware, yet an advertisement or false download button may lead the user to an unsafe site. The United States Federal Trade Commission therefore advises users to avoid software-download advertisements and unfamiliar download sites and to obtain software from a source they know and trust [8].

Paid software often has funding for continued development, documentation, technical support, testing, and regular updates. An established developer also has a reputation and customer relationships to protect. These considerations can provide useful evidence of trustworthiness,

but payment is not a guarantee of quality, security, or privacy. A paid product can contain a vulnerability, stop receiving updates, impose restrictive licence conditions, collect more data than the task requires, or show advertising. Applications and websites may use identifiers and other information for tracking and personalised advertising regardless of whether the user paid for access [9].

Software	Possible advantages	Questions to ask
Freeware	No purchase cost; easy to try; may be sufficient for a small or occasional task.	Is this the developer's official copy? How is development funded? Does it display advertisements, install extras, collect data, receive updates, and provide a way to export files?
Paid software	May provide sustained development, documentation, support, updates, and specialised features.	Is the developer dependable? What is the total cost? What happens when payment or support ends? What data are collected, and can work be moved to another product?

Price should therefore be treated as one piece of evidence rather than a security test. Before installing any program, confirm the developer and exact website or official store, avoid advertisements masquerading as download buttons, check the requested permissions, and keep both the program and device updated. For important work, also confirm that files can be backed up and exported. These checks apply to free, paid, subscription, and open-source software alike.

Practical Work

List the digital systems you have used since waking up. Choose one activity and trace its input, processing, storage, output, and communication. Identify the hardware, software, data, and networks involved. For one piece of software, identify its principal use and how it is supplied or paid for. End by stating one thing about the system that you would like this course to help you understand.

1.5 Questions for Discussion and Exercises

These questions form a bank from which an instructor may select. They are not intended as a single sitting's checklist.

1. A student enters observations into an online form during a biology laboratory activity. Identify a possible input, processing step, form of storage, output, and act of communication. Name the main hardware and software involved.
2. Explain the difference between RAM and long-term storage. Why would a computer need both?

3. A laptop is advertised as having more CPU and GPU cores than another model. Explain why those figures alone do not prove that it will be faster for a student's work. Identify one non-upgradeable component and explain how the words "up to" should affect the interpretation of its battery claim.
4. The supplied laptop sheet gives two memory-bandwidth figures. Explain how the manufacturer's specification resolves the conflict and what this example teaches about apparently authoritative tables.
5. Why should a buyer check the exact regional model of a phone rather than relying on its product name alone? In your answer, use the SIM or battery differences in Table 1.2, and explain why three 48-megapixel cameras are not one 144-megapixel camera.
6. Explain the difference between the connection that brings internet service into a home and the Wi-Fi network inside it. How could you investigate which one is causing a slow connection?
7. Ignoring overhead, estimate how long a 100 Mbps connection would take to download an 800 MB file. Show the conversion between bytes and bits before calculating the time.
8. A household regularly makes video calls, uploads large files, and has several simultaneous users. Apart from advertised download rate, what features and plan conditions should it examine before subscribing?
9. Explain why SaaS, subscription software, freeware, and open-source software are not four mutually exclusive categories. Then assess the claim that freeware is probably malware whereas paid software is safe.
10. A biology group needs software for recording observations, analysing data, and sharing results. State the questions the group should ask before choosing a product. Include compatibility, cost, privacy, data export, and continued access in your answer.

Remaining Chapters Under Construction

This edition contains the Introduction and Chapter 1. Further chapters are being prepared and will be added as the course progresses. Their contents may also change in response to students' questions, suggestions, and practical needs.

Appendix A

Sample Answers to Exercises

The answers in this appendix are examples, not the only acceptable responses. Several questions allow more than one answer because the circumstances, priorities, or equipment may differ. An alternative answer may be equally good when it is accurate, relevant to the question, and clearly justified.

A.1 Chapter 1: How IT Works and Why It Matters

1. The observations entered by the student are the input. The online form may check whether required fields have been completed or convert an entered date into a standard format; this is processing. The responses may be stored in a remote database or spreadsheet. A confirmation message or summary table is an output. Sending the responses from the device to the remote system is an act of communication. The hardware may include a phone or laptop, router, network equipment, and server. The software may include the operating system, browser, online form, and database.
2. RAM holds data and instructions that programs are using at the moment. It is fast working space, and its contents are normally lost when the device is switched off. Long-term storage, such as an SSD, retains applications and files when power is removed. A computer needs RAM for active work and storage for keeping work between sessions.
3. Core counts cannot be compared in isolation because processor design, generation, cooling, software, and the kind of work all affect performance. Suitable benchmarks or tests of the student's actual applications would provide better evidence. In the example laptop, the RAM and internal SSD cannot be upgraded after purchase. The words *up to* indicate a best-case battery result under stated test conditions, not the running time that every user will obtain.
4. Apple's specification assigns 614 GB/s to the M5 Max configuration with a 40-core GPU and 460 GB/s to the configuration with a 32-core GPU. The apparent conflict arose because the supplied

sheet presented figures for different configurations. The example shows that a detailed table can still be ambiguous or wrong. Important specifications should be checked against a primary source and the exact model being considered.

5. Regional models may differ in SIM arrangement, battery capacity, supported network bands, and warranty coverage. In the table, some versions accept a physical Nano-SIM while another is eSIM-only, and the listed battery capacities also differ. The exact model therefore matters even when the product name is unchanged. The three rear cameras perform different optical roles. Their 48-megapixel figures cannot be added together as though they formed one 144-megapixel camera.
6. The provider's connection carries data between the home and the wider internet. Wi-Fi distributes that connection wirelessly within the home. To investigate a slow connection, test a computer through Ethernet and compare the result with Wi-Fi tests made near and far from the router. A good Ethernet result but poor distant Wi-Fi result suggests an internal wireless problem. Poor results on both may point to the service, modem or router, congestion, or another shared-network problem.
7. First convert the file size from megabytes to megabits:

$$800 \text{ MB} \times 8 = 6,400 \text{ Mb.}$$

The estimated download time is therefore

$$\frac{6,400 \text{ Mb}}{100 \text{ Mb/s}} = 64 \text{ s.}$$

The actual transfer would usually take longer because of overhead, congestion, Wi-Fi conditions, and other limitations.

8. The household should examine upload rate, latency, jitter, packet loss, and performance during busy periods. It should also check whether the advertised rate is a maximum or expected range, whether capacity is shared, and whether the plan has a data allowance or fair-use threshold. Other relevant matters include availability at the exact address, installation and equipment charges, contract length, later price increases, reliability, and repair response. The importance assigned to each factor may vary among households.
9. SaaS describes how software is delivered; subscription describes how access is paid for; freeware describes a zero purchase price; and open source describes licence permissions. One product can fit several of these descriptions at once. The claim that freeware is probably malware while paid software is safe is too broad. Either can be secure or unsafe. Better evidence includes

the identity and reputation of the developer, the official download source, permissions, update history, privacy practices, independent reviews, and the ability to export files.

10. The group should begin by defining what must be recorded, which analyses are required, and how results will be shared. It should check compatibility with members' devices and required file formats; the full cost over the project; ease of use, documentation, and support; offline access; backup and recovery; and whether data can be exported in a usable form. Because biological work may involve personal, health, location, or unpublished research data, the group should also examine permissions, storage location, access controls, encryption, privacy terms, and what happens to the files when payment or institutional access ends.

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